Sample space, outcomes, events

Outline

Topics

- Review of basic probability theory concepts: outcome, event, sample space
- Intuition from the Bayesian perspective

Rationale

- One definition of Bayesian inference: applying *probability theory* to *statistical inference problems*
  - Therefore, it is critical to understand probability to learn Bayesian inference
  - This week, we will help you "reload in memory" some of the most important bits of probability theory used in this course

$x = 0, 1, 2$
number of coins

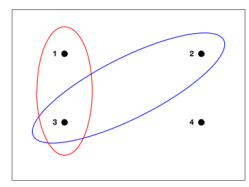
4 flips

outcome of #1: $S = (x, y_1, y_2, y_3, y_4)$

value of ith flip.

Definitions

- **Sample space**, denoted S, a set.
  - Example: $S = \{1, 2, 3, 4\}$ (see Figure).
- Each element s of S is called an **outcome**, $s \in S$.
  - Example: each of the 4 points.
- A set of outcomes $E \subset S$ is called an **event**.
  - Example: $E = \{s \in S : s \text{ is odd}\}$ (red in the Figure).



Intuition: Bayesian view

- In Bayesian statistics, an outcome will describe the state of the world.
- We do not know which outcome is the true state of the world.
- We **observe** partial information on the state of the world/outcome.
- We rule out the outcomes that are not consistent with the observation…
- …but there will be several outcomes left!
  - We will deal with this situation using probability theory.

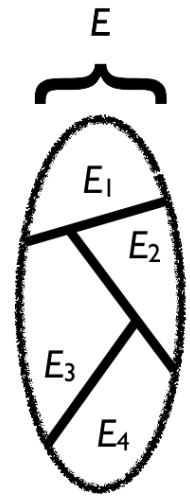
Intuition: randomized algorithms

- An algorithm is "randomized" if it has access to virtual dices/coins.
- In practice this is done using pseudorandom number generators.
- In this context an outcome is a random seed, i.e. the initialization of

the pseudorandom number generator.

Axioms of probability

Partitions



Definition: $E_1, E_2, \ldots$ is a **partition** of E if:

- the E_i's are disjoint, i.e.,

$$E_i \cap E_j = \emptyset \text{ when } i \neq j,$$

- and their union is E, i.e., $\cup_i E_i = E$.

Axioms of probability

- A **probability** is a function $\mathbb{P}$ that satisfy the following constraints:
  - $\mathbb{P}$ should take events as input and return a number between zero and one:

$$\mathbb{P}(E) \in [0, 1].$$

  - *Additivity axiom*: if $E_1, E_2, \ldots$ is a partition of E, then

$$\mathbb{P}(E) = \sum_i \mathbb{P}(E_i).$$

  - $\mathbb{P}(S) = 1$
- Thanks to the constraints, even if I only specify a few known probabilities I can recover many other ones mathematically/computationally.

Random variables

Outline

Topics

- Random variable as mathematical objects.
- Notation convention for observation/latent

Rationale

Random variables are used as building blocks for two key uses in Bayesian stats: modelling "knowns" (observations) and "unknowns" (latent variables/parameters/prediction).

unobserved/unknown.

Definition

A (real) **random variable** is a function from a sample space S to the reals, $X : S \to \mathbb{R}$.

Example:

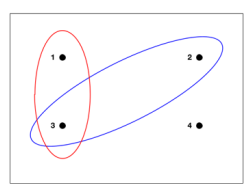
- Continuing the example with $S = \{1, 2, 3, 4\}$.
- Consider $X(s) = 1$ if s is odd, and $X(s) = 0$ otherwise.

Probabilist's notation

- Let X denote a random variable.
- The notation $(X = 1)$ or $(X \in E)$ is invalid in set theory.
- Therefore, probabilists "gave it a meaning" as follows:

$$(X = 1) = \{s : X(s) = 1\}.$$

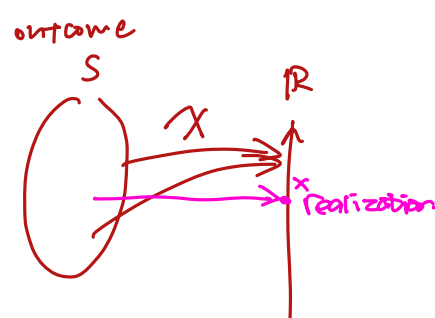
Example: Consider $X(s) = 1$ if s is odd, and $X(s) = 0$ otherwise. Then $(X = 1)$ corresponds to the red circle.



Conventions: probability vs Bayesian

Probability convention:

- Random variables are denoted with capitals in probability theory
- The same letter in small cap is used for a dummy variable holding the output of the random variable.
  - Note: "A dummy variable holding the output of the random variable" is called a **realization**.



variable" is called a **realization**.
    - Example: X for the random variable and x for its realization.
- We will start off using this convention in the first few weeks.

Bayesian statistics convention:

- Often the capitalization convention is not used in the Bayesian statistics literature.
- Hence we will eventually drop the probability theory capitalization convention.

More conventions

- X: unobserved random variable (synonym of "unobserved": **latent**)
- Y: observed random variable

More precisely:

- Y is the "mechanism of observation"..
- whereas the actual observation is a realization y of Y.

Extension

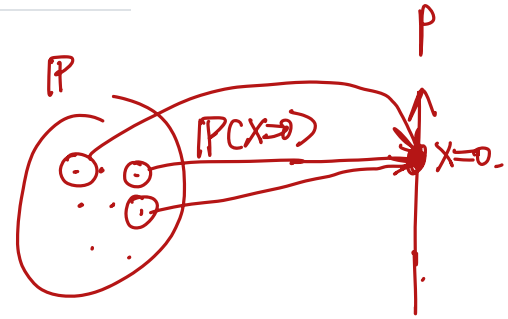
A **random vector** is a function from a sample space to $\mathbb{R}^n$.

Example in Bayesian statistics: the vector (X, Y) containing both the unobserved and observed quantities.

Probability mass functions

Definition

Probability Mass Function (PMF):

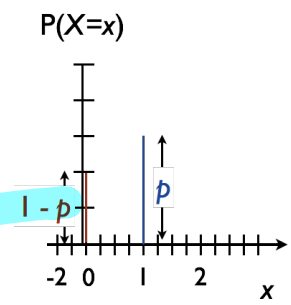


- denoted by p (not to be confused by $\mathbb{P}$),
- defined by:

$$p(x) = \mathbb{P}(X = x).$$

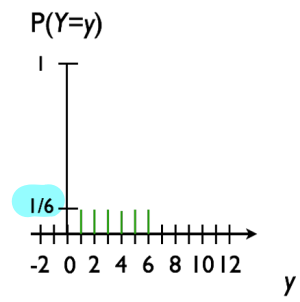
- Terminology: $\{x : p(x) > 0\}$ is called the **support**.
- If there are several random variables, we use a subscript to disambiguate the PMFS, e.g.,
  - p_X for the PMF of the latent random variable,
  - p_Y for the PMF of the observed random variable.

Example: coins



- PMF of fair coin: $p(0) = p(1) = 1/2$
- Generalization: the family of **Bernoulli PMFs**
- Recall: when you flip a "Biased" coin
  - get a 1 with probability $p \in [0, 1]$
  - get a 0 with probability $1 - p$
- p is an example of what is called a PMF's *parameter*
- Notation: $X \sim \text{Bern}(p)$

Example: discrete uniform distributions



- Example: dice
- Notation: $Y \sim \text{Unif}\{1, 2, 3, 4, 5, 6\}$

Simulation/sampling from a PMF

In R

```
require(extraDistr)
```

```
Loading required package: extraDistr
```

```
# 10 coin flips:
```

```
rbern(10, prob=0.5)
```

```
[1] 1 1 0 1 1 0 0 0 1 1
```

```
# two dice rolls
rdunif(2, min=1, max=6)
```

```
[1] 3 6
```

How does it work?

- Think of the green lines in the uniform PMF as "sticks" with "labels."
  - The "labels" are the different realization, e.g. 1, 2, 3, …, 6.
- Place the six sticks in the interval $[0, 1]$ so that they do not overlap.
- Sample a uniform real number in $[0, 1]$
  - in R: `runif(1)`
- The uniform falls in exactly one of the sticks.
- Return the label of that stick.

Mathematical explanation

The above "stick-based" algorithm can be implemented using the cumulative distribution function and a generalization of its inverse known as the quantile function.

Cumulative distribution function (CDF): $F(x) = \mathbb{P}(X \leq x)$.

Quantile function: $Q(u) = \inf\{x \in \mathbb{R} : u \leq F(x)\}$.

Then the "stick-based" algorithm can be written as:

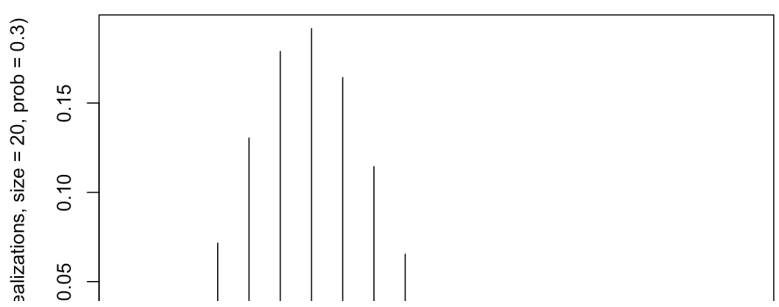
1. $U \sim \text{Unif}[0, 1]$
2. Return $Q(U)$.

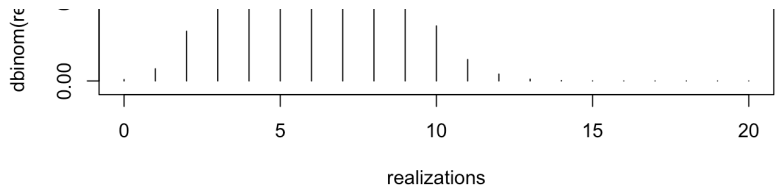
Plotting PMFs in R

Here is an example of how to plot PMFs in R:

```
realizations <- 0:20

plot(realizations, dbinom(realizations, size=20,
```





Some explanations:

- Here `dbinom` is the R function for the PMF of a binomial (R uses the prefix `d___` for densities and PMFs).
- We use the fact that many functions in R such as this one are vectorized, i.e. the normal PMF function takes a single point x and output $p(x)$ where p is the PMF; the vectorized version takes vector $(x_1, x_2, \ldots)$ and returns $(p(x_1), p(x_2), \ldots)$.
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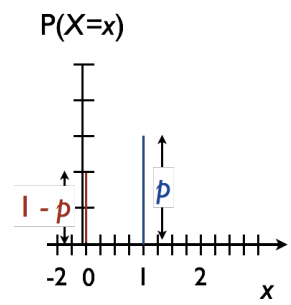

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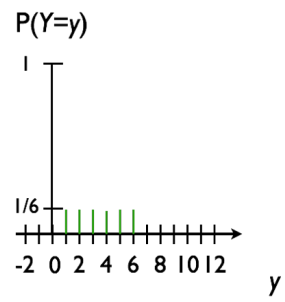

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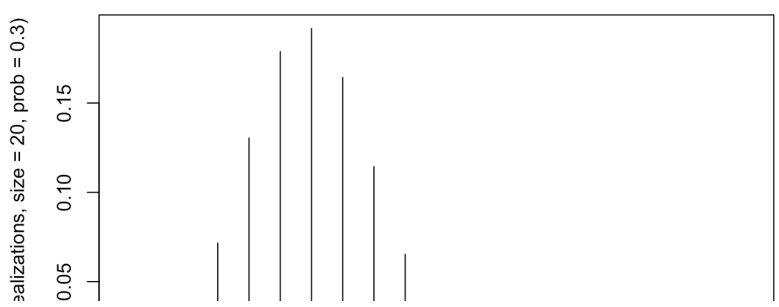

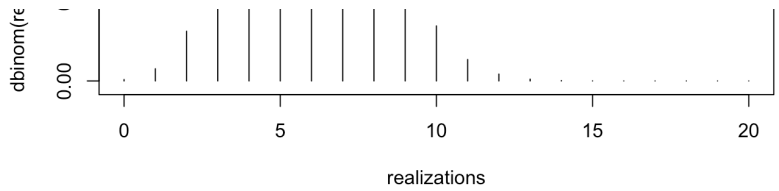

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Decision trees

Outline

Topics

- Decision trees
- Review of more probability theory concepts, contextualized in decision trees: outcome, event, sample space, partitions, conditional probability

Rationale

We will use decision trees to provide visualization for a bunch of complex concepts such as forward simulation, posterior inference, importance sampling, probabilistic programming, etc.

Running example

- Imagine a bag with 3 coins each with a different probability parameter p
- Coin $i \in \{0, 1, 2\}$ has bias $i/2$—in other words:
  - First coin: bias is $0/2 = 0$ (i.e. both sides are "heads", $p = 0$)
  - Second coin: bias is $1/2 = 0.5$ (i.e. standard coin, $p = 1/2$)
  - Third coin: bias is $2/2 = 1$ (i.e. both sides are "tails", $p = 1$)

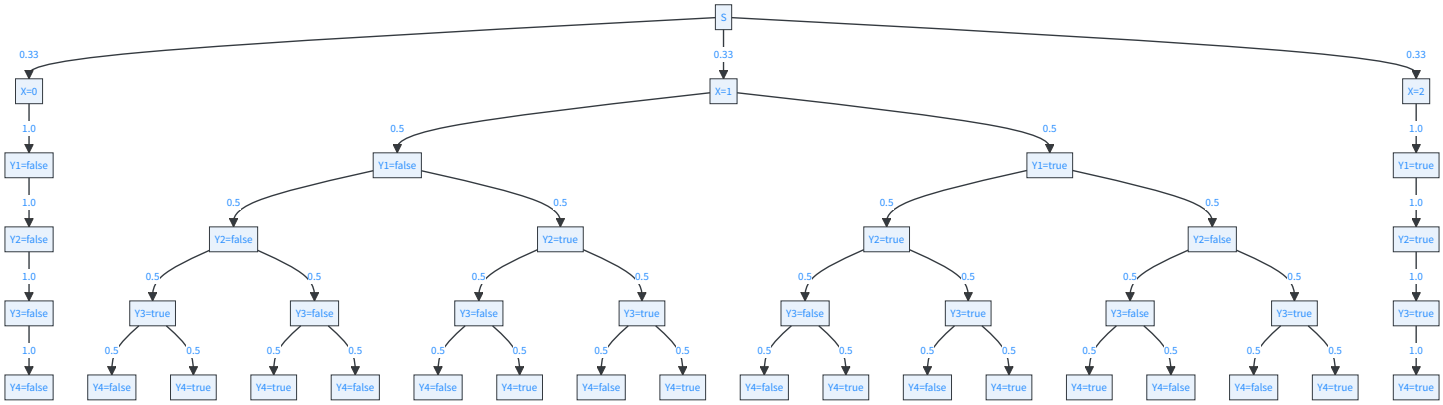
- Consider the following two steps sampling process
  - Step 1: pick one of the three coins, but do not look at it!
  - Step 2: flip the coin 4 times
- Mathematically, this probability model can be written as follows:

$$\begin{aligned} X &\sim \text{Unif}\{0, 1, 2\} \\ Y_i | X &\sim \text{Bern}(X/2) \end{aligned} \tag{1}$$

Decision tree

- **Decision tree**: a recursive classification of all possible scenarios
- Nodes in the tree are "groups of scenarios" which we call **events**
- Children of a node partitions an event into an exhaustive set of sub-cases,
  - i.e. $E_1, E_2, \ldots$ is a **partition** of E.
- In the decision tree below, we partitioned events until we get events at the leaves each containing a single scenario
  - We call one individual scenario an **outcome**
  - We call the set of all outcomes the **sample space**, S, and put it at the root of decision trees

at the root of decision trees.



Nodes and events

To describe the event corresponding to a node v in the tree:

1. trace the path from the node v to the root
2. take the intersection of all node labels.

Example: find the node in the above tree corresponding to the event $(X = 1) \cap (Y_1 = 0)$.

Probability notation review:

- $(X = 1) = \{s \in S : X(s) = 1\}$
- $(X = 1, Y_1 = 0) = (X = 1) \cap (Y_1 = 0)$

Edges and conditional probabilities

When there is an edge from events E_1 to E_2, we annotate it with $\mathbb{P}(E_2|E_1)$.

Recall: conditional probability of E_2 given E_1

$$\mathbb{P}(E_2|E_1) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_1)}$$

Example:

- take the edge from $E_1 = (X = 1)$ to $E_2 = (X = 1, Y_1 = 0)$.

$$\mathbb{P}(E_2|E_1) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_1)} = \frac{\mathbb{P}(E_2)}{\mathbb{P}(E_1)} = \mathbb{P}(Y_1 = 0|X = 1)$$

- Translating $\mathbb{P}(Y_1 = 0|X = 1)$ into words: "the probability that the first flip is 'heads' $(Y_1 = 0)$ given that you picked the standard coin $(X = 1)$."
- Hence the edge from E_1 to E_2 is labelled $1/2$.

Forward sampling

Outline

Topics

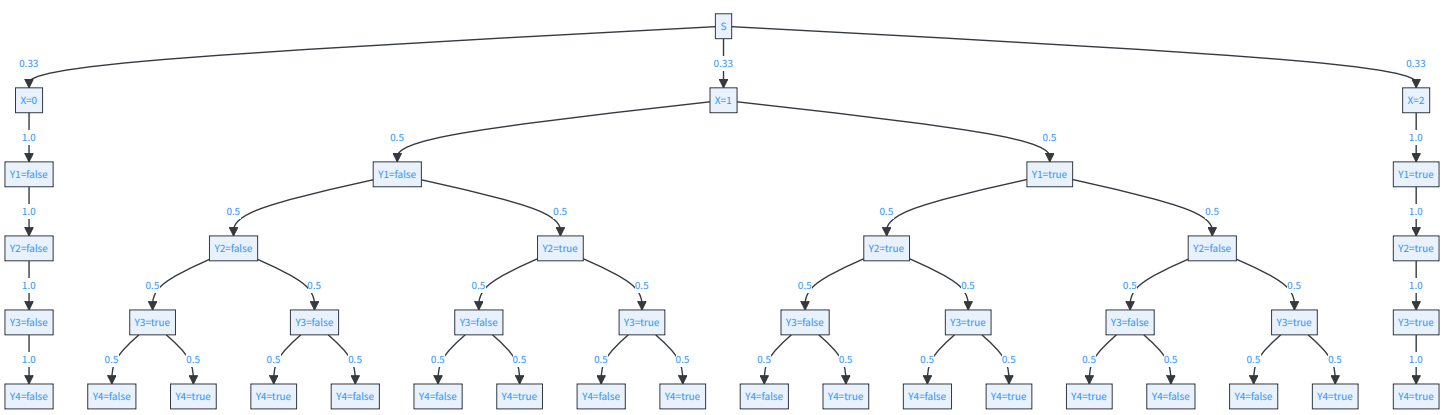
- Notion of forward sampling (also known as forward simulation)
- How to do it in practice
  - Useful functions
  - Graphical models

Rationale

- Sampling is the main way Bayesian inference is performed nowadays.
- We introduce here the simplest flavour of sampling, *forward sampling*.
- Bayesian inference mostly uses a more complicated type of sampling called *posterior sampling* which we will cover later.
- But forward sampling is still helpful to help debug Bayesian inference software as we will see soon.

Forward sampling as depth-first traversal

Recall our recurring bag sampling example, with its corresponding decision tree:



- Forward simulation is a type of tree traversal. I.e. moving from node to node in the tree.
- Forward simulation is a recursive process initialized at the root of the decision tree (labelled S).
  - When we are a node v in the tree, we pick one of v's children at random.
    - More precisely, we use methods discussed in the previous section on simulation from PMFs.

section on simulation from PMFs
    - We recurse until we reach a leaf.
        - From this leaf we obtain an outcome and hence a realization for all random variables, both "observed" and "unobserved."

Forward sampling as specifying a model

We have encountered that notation earlier:

$$X \sim \text{Unif}\{0, 1, 2\}$$
$$Y_i | X \sim \text{Bern}(X/2).$$

- This notation is a recipe providing all the information required to perform forward sampling.
    - Specifically, the PMF to use at each recursion step.
    - In continuous models, it will be the same idea except that we will have a probability density instead of a PMF.

Example 1

You will practice forward sampling in Ex1, Q1.1.2

Example 2

Consider the following model:

$$X \sim \text{Unif}\{1, 2, 3, 4\}$$
$$Y | X \sim \text{Unif}\{1, \ldots, X\}.$$

Interpretation: you roll a D&D d4 dice (blue one on the image), then pick an integer uniformly between 1 and the number on the dice.

Example of forward simulation code:

```
require(extraDistr)
```

```
Loading required package: extraDistr
```

```
set.seed(4)

forward_simulate_roll_and_pick <- function() {
  x <- rdunif(1, min=1, max=4) # 1 uniform betwee
  y <- rdunif(1, min=1, max=x)
  c(x, y) # return a vector with these two realiz
}
```

```
forward_simulate_roll_and_pick()
```

```
[1] 3 1
```

```
forward_simulate_roll_and_pick()
```

```
[1] 2 1
```

```
forward_simulate_roll_and_pick()
```

```
[1] 4 2
```

Conditioning

Outline

Topics

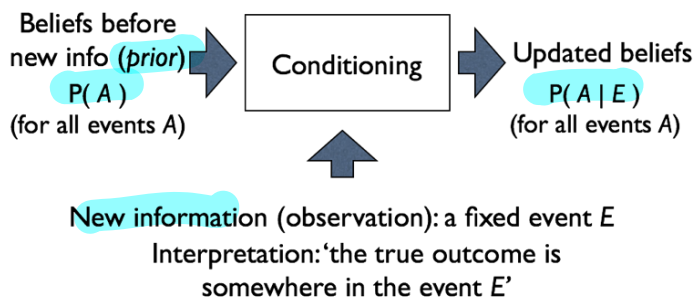
- Intuition on conditioning
- A conditional probability is a probability.

Rationale

- Conditioning is the workhorse of Bayesian inference!
  - Used to define models (as when we assigned probabilities to edges of a decision tree)
  - And soon, to gain information on latent variables given observations.

Conditioning as belief update

Key concept: Bayesian methods use probabilities to encode *beliefs*.

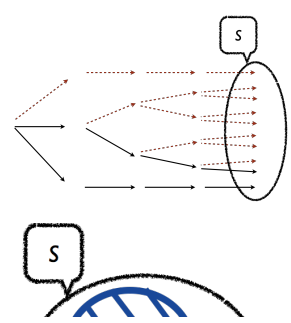


We will explore this perspective in much more details next week.

A conditional probability is a probability

The "updated belief" interpretation highlights the fact that we want the result of the conditioning procedure, $\mathbb{P}(\cdot|E)$ to be a probability when viewed as a function of the first argument for any fixed E.

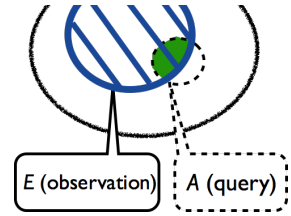
Intuition behind conditioning



- For a query even A, what should be the updated probability?
- We want to remove from A all the outcomes that are not compatible with the new information E. How?
  - Take the intersection: $A \cap E$

- We also want: $\mathbb{P}(S|E) = 1$ (last section)
- How? *Renormalize*:[1]

$$\mathbb{P}(A|E) = \frac{\mathbb{P}(A, E)}{\mathbb{P}(E)}$$



Footnotes

1. In the equation, we use comma to denote intersection, i.e., $(A \cap E) = (A, E)$. ↩

Chain rule

Outline

Topics

- Mathematical statement
- Visual intuition on a decision tree
- Special names for the pieces of chain rule (joint and conditional PMFs)
- Conditional independence

Rationale

The chain rule allows us to compute a probability that the forward sampling function takes a given path.

The chain rule seem innocent but is used heavily in Bayesian statistics. It is also the building block for Bayes rule.

Proposition

If E_1 and E_2 are any events,

$$\frac{P(E_2, E_1)}{P(E_1)}$$

$$\mathbb{P}(E_1, E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2|E_1).$$

This is true in any order, i.e. we also have $\mathbb{P}(E_1, E_2) = \mathbb{P}(E_2)\mathbb{P}(E_1|E_2)$.

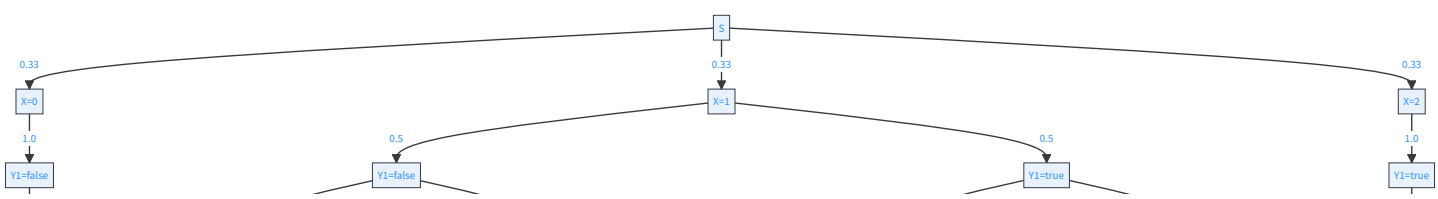
Generalization

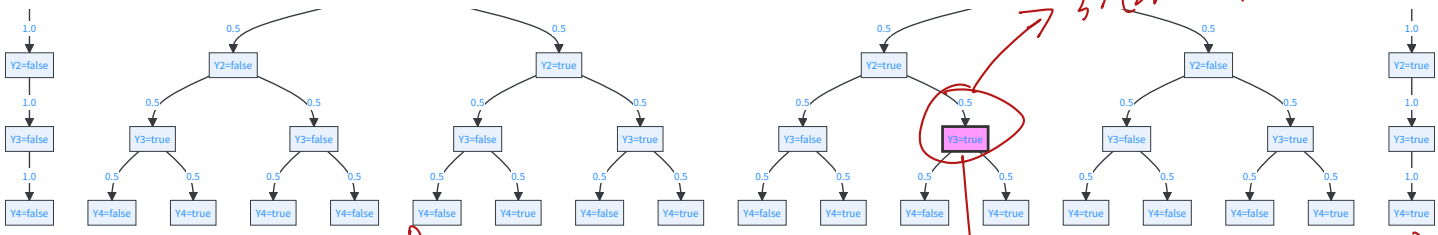
For any events $E_1, E_2, E_3 \ldots$, No assumptions ⇒ Always true.

$$\mathbb{P}(E_1, E_2, E_3 \ldots) = \mathbb{P}(E_1)\mathbb{P}(E_2|E_1)\mathbb{P}(E_3|E_1, E_2)\ldots.$$

Visual intuition

Chain rule: the probability of a node is the product of the edge labels on the path to the root.





$\frac{1}{2} \times (\frac{1}{2})^3 = \frac{1}{64}$.

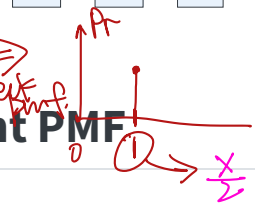
$\rightarrow P(Y_3=1, Y_2=1, Y_1=1, X=1)$

$= P(X=1)\, P(Y_1=1 \mid X=1)\, P(Y_2=1 \mid X=1, Y_1=1)$

$P(Y_3=1 \mid X=1, Y_1=1, Y_2=1)$.

Chain rule

$Y_i \mid X \sim \text{Bern}(\frac{X}{2}) \Rightarrow$ check pmf.



$\rightarrow \frac{X}{2} \Rightarrow X=2,$

Joint PMF

We will often encounter expression of the form of a conjunction (intersection/and) of several variables. A handy notation for that is the **joint PMF**

For example, here is the joint PMF of (X, Y_1, Y_2, Y_3):

$$p(x, y_1, y_2, y_3) = \mathbb{P}(X = x, Y_1 = y_1, Y_2 = y_2, Y_3 = y_3).$$

Sometimes we put the random variables in question as subscript, for example $p_{X,Y_1}(x, y)$ for the joint PMF of X and Y_1.

Conditional PMF

Similarly, here is an example of a conditional PMF:

$$p_{Y_1|X}(y|x) = \mathbb{P}(Y_1 = y|X = x).$$

Conditional independence

The model was specified as:

$$X \sim \text{Unif}\{0, 1, 2\}$$
$$Y_i|X \sim \text{Bern}(X/2)$$

i.e. with $\mathbb{P}(X = x)$ and $\mathbb{P}(Y_i = y|X = x)$ for all x and y.

Question: how did we go from $\mathbb{P}(Y_2 = 1|X = 1, Y_1 = 1)$ (in our chain rule computation) to $\mathbb{P}(Y_2 = 1|X = 1)$ (model specification)?

Definition: V and W are conditionally independence given Z if

$$\mathbb{P}(V = v, W = w|Z = z) = \mathbb{P}(V = v|Z = z)\mathbb{P}(W = w|Z = z).$$

~~**Exercise:**~~ show the above definition is equivalent to:

$$\mathbb{P}(V = v|W = w, Z = z) = \mathbb{P}(V = v|Z = z).$$

X: unknown param. Y: data observed.

$P(X=x|Y=y) = \frac{P(X=x \cap Y=y)}{P(Y=y)} \rightarrow P(Y=y|X=x)P(X=x)$

$\sum_x P(Y=y|X=x)P(X=x)$

所有导致Y=y的情况相加.

Bayes rule

Outline

Topics

- Bayes rule for discrete models
- Visual intuition

Rationale

First example of computing a *posterior distribution*, a key concept in Bayesian statistics.

Running example

- Imagine a bag with 3 coins each with a different probability parameter p
- Coin $i \in \{0, 1, 2\}$ has bias $i/2$—in other words:
  - First coin: bias is $0/2 = 0$ (i.e. both sides are "heads", $p = 0$)
  - Second coin: bias is $1/2 = 0.5$ (i.e. standard coin, $p = 1/2$)
  - Third coin: bias is $2/2 = 1$ (i.e. both sides are "tails", $p = 1$)

- Consider the following two steps sampling process
  - Step 1: pick one of the three coins, but do not look at it!
  - Step 2: flip the coin 4 times
- Mathematically, this probability model can be written as follows:

$$\begin{aligned} X &\sim \text{Unif}\{0, 1, 2\} \\ Y_i|X &\sim \text{Bern}(X/2) \end{aligned} \tag{1}$$

Consider the second question in the first exercise:

Suppose now that you observe the outcome of the 4 coin flips, but not the type of coin that was picked. Say you observe: "heads", "heads", "heads", "heads" = `[0, 0, 0, 0]`. Given that observation, what is the probability that you picked the standard coin (i.e., the one with $p = 1/2$)?

Strategy

Denote the observation by $y_{1:4} = (0, 0, 0, 0)$. In the rest of the argument we will always fix y to that value.

1. Attack the more general problem $\pi(x) = \mathbb{P}(X = x|Y_{1:4} = y_{1:4})$ for **all** hypotheses $x \in \{0, 1, 2\}$ instead of just the requested $x = 1$ (corresponding to the "standard coin").
2. By definition of conditioning:

$$\pi(x) = \frac{\mathbb{P}(X = x, Y_{1:4} = y_{1:4})}{\mathbb{P}(Y_{1:4} = y_{1:4})}.$$

Let us call the numerator joint pmf.

$$\gamma(x) = \mathbb{P}(X = x, Y_{1:4} = y_{1:4}),$$

and the denominator,

$$Z = \mathbb{P}(Y_{1:4} = y_{1:4}).$$

3. Start by computing $\gamma(x)$ for all x. (using chain rule)
4. Note $Z = \gamma(0) + \gamma(1) + \gamma(2)$ (why?).

$x \in \{0,1,2\}$

$P(X=0) = P(X=1) = P(X=2) = 1/3.$

#3 $\gamma(x) = P(Y_{1:4} = y_{1:4} | X = x) P(X = x)$

$= \frac{1}{3}(1 + \frac{1}{16} + 0) = \frac{1}{3} \cdot \frac{17}{16} = \frac{17}{48}.$

$P(y_1, y_2, y_3, y_4 | X = 0,1,2)$

$= \sum P(y_{1:4} | X = x)$

① $P(y_{1:4} | X=0) = 1^4 = 1$

② $P(y_{1:4} | X=1) = (\frac{1}{2})^4 = 1/16$

③ $P(y_{1:4} | X=2) = 0$

#4 $Z = P(Y) = \sum P(Y|X) P(X)$

$= \gamma(0) + \gamma(1) + \gamma(2)$

Expectations

Outline

Topics

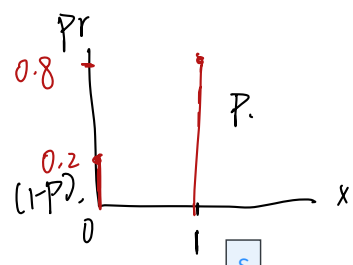
- Expectation for discrete random models
- Law of the Unconscious Statistician

Rationale

Expectation si the main tool to translate a posterior distribution into the various outputs of Bayesian inference (point estimate, credible intervals, prediction, action).

Expectation of a single random variable

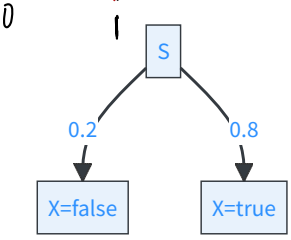
$E(X) = 0 \cdot 0.2 + 1 \cdot 0.8 = 0.8$



Recall:

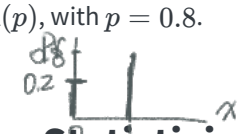
$$\mathbb{E}[X] = \sum_x x p_X(x),$$

where the sum is over the point masses of X, i.e. $\{x : p_X(x) > 0\}$.



Example: compute $\mathbb{E}[X]$ if $X \sim \mathrm{Bern}(p)$, with $p = 0.8$.

$E(X) = 1 \times 0.8 + 0 \cdot 0.2 = 0.8$



Law of the Unconscious Statistician

Proposition: if g is some function,

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).$$

$\Rightarrow E(X^2) = \sum x^2 p_X(x).$

Example: compute $\mathbb{E}[X^2]$ if $X \sim \mathrm{Bern}(p)$, and hence $\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

$\mathrm{Var}(X|E) = E(X^2|E) - (E(X|E))^2$

\# $X \sim \mathrm{Bern}(1/3)$

$\Rightarrow E(1/(X+1)) = \frac{1}{0+1} \cdot \frac{2}{3} + \frac{1}{1+1} \cdot \frac{1}{3} = \frac{2}{3} + \frac{1}{6}$

$= \frac{5}{6}.$

Expectation of a function of several random variables

$E(1/(X+1)) = \sum \frac{1}{X+1} \cdot P(X)$

$= \frac{1}{0+1} \cdot \frac{2}{3} + \frac{1}{1+1} \cdot \frac{1}{3}$

Let us go back to our running example:

- Imagine a bag with 3 coins each with a different probability parameter p
- Coin $i \in \{0, 1, 2\}$ has bias $i/2$—in other words:

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- Consider the following two steps sampling process
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- Mathematically, this probability model can be written as follows:

$$\begin{aligned} X &\sim \text{Unif}\{0, 1, 2\} \\ Y_i | X &\sim \text{Bern}(X/2) \end{aligned} \tag{1}$$

Example: computing $\mathbb{E}[X(Y_1 + 1)]$ (similar to what you will be doing in the exercise in question 1.1)

$g(x, y_{1:4}) = x(y_1+1)$.

Note: this is of the form $\mathbb{E}[g(\ldots)]$, so we can use the Law of the Unconscious Statistician.

$\Rightarrow E[X(Y_1+1)] = E[g(X, y_{1:4})]$

How to do it:

for $x=2$, $g(2, y_{1:4}) = 2(1+1) = 4$.
$P(2, y_{1:4}) = \frac{1}{3} \times 1 \times 1 \times 1 \times 1 = 1/3$ > 相乘.

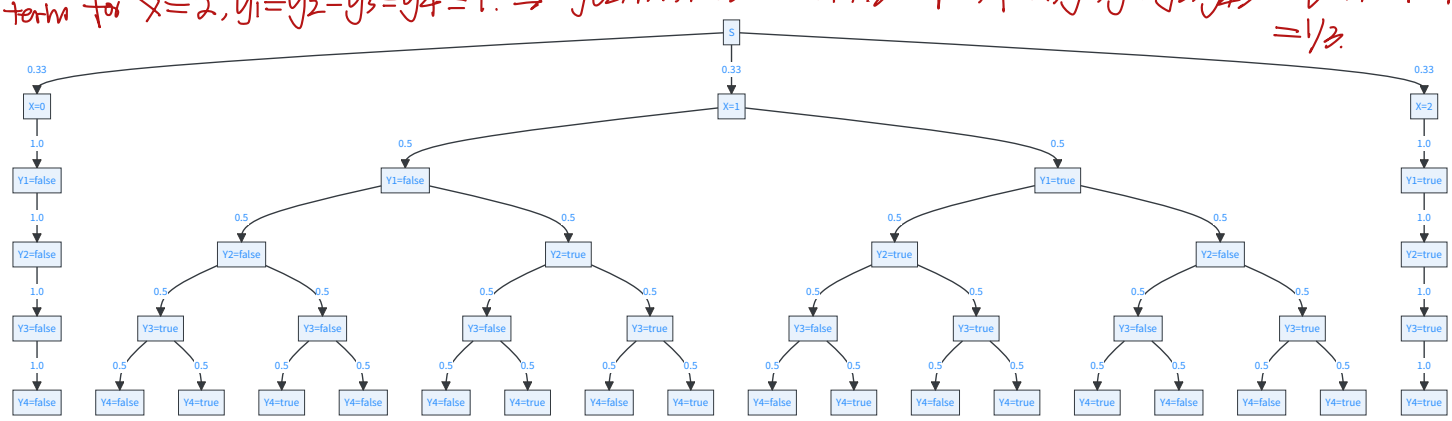
- first, identify g, here it is $g(x, y_1, \ldots, y_4) = x(y_1 + 1)$ (in the exercise it is slightly different)
- denote by p the joint PMF of **all** the random variables in the model
- compute the expectation using

$$\mathbb{E}[g(X, Y_1, \ldots, Y_4)] = \sum_x \sum_{y_1} \sum_{y_2} \ldots \sum_{y_4} g(x, y_1, \ldots, y_4) p(x, y_1, y_2, y_3, y_4).$$

- Each sum runs over the point mass of its PMF as before, e.g. $x \in \{0, 1, 2\}$.
- Recall: $p(x, y_1, y_2, y_3, y_4)$ can be computed using the chain rule.

Recall the decision tree, how to visualize the above equation?

last term for $x=2, y_1=y_2=y_3=y_4=1$. $\Rightarrow g(2,1,1,1,1) = 2(1+1) = 4$, $P(x, y_1, y_2, y_3, y_4) = \frac{1}{3} \times 1 \times 1 \times 1 \times 1 = 1/3$.



Bayes rule

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and the denominator,

$$Z = \mathbb{P}(Y_{1:4} = y_{1:4}).$$

3. Start by computing $\gamma(x)$ for all x. (using chain rule)
4. Note $Z = \gamma(0) + \gamma(1) + \gamma(2)$ (why?).

$E(X) = \sum x P(x)$.

Simple Monte Carlo ⇒ 不特别有可能性计算，按prob.随机方法抽样再算Alg.

Outline

Topics

- Simple Monte Carlo method
- Theoretical guarantee from the Law of Large Numbers

Rationale

Simple Monte Carlo is the foundation for more complex Monte Carlo methods used by Bayesian practioners (e.g. Importance Sampling and Markov chain Monte Carlo (MCMC))

Approximation of expectations using forward simulation ①随机生成 ②满足条件次下生成.

- As before, we want to compute an expectation $\mathbb{E}[g(X, Y)]$
- Imagine a very large decision tree, but where most branches have very low probability and only few have large probability
- In this case, instead of computing the exact expectation by iterating over each of the leaves as before, we will *approximate* expectations using forward simulation (a method know as **simple Monte Carlo**)
- This is done as follows:
  - Call your forward simulator M times. 把"生成一对(X,Y)"重复M次".
    - Denote the output at iteration $m \in \{1, 2, \ldots M\}$ by:

$$(X^{(m)}, Y^{(m)}) \sim p_{X,Y}(\cdot)$$

    - Compute g on each, call each of the M outputs $G^{(m)}$ ⇒ $G^{(1)}, \ldots, G^{(M)}$.

func.

$$G^{(m)} = g(X^{(m)}, Y^{(m)}).$$

  - Return the average

$$\hat{G}_M = \frac{1}{M} \sum_{m=1}^{M} G^{(m)}.$$

Intuitively, the output $\hat{G}_M$ has the nice property:

$$\hat{G}_M \approx \mathbb{E}[g(X, Y)]. \quad \text{since LLN.} \tag{1}$$

Example 1

Question 1.2 in the exercise.

Example 2

Consider the following model:

$$X \sim \text{Unif}\{1, 2, 3, 4\}$$
$$Y|X \sim \text{Unif}\{1, \ldots, X\}.$$

Interpretation: you roll a D&D d4 dice (blue one on the image), then pick an integer uniformly between 1 and the number on the dice.

Let us use Monte Carlo to estimate $\mathbb{E}[X^Y]$:

```
require(extraDistr)
```

```
Loading required package: extraDistr
```

```
set.seed(4)

# Recall our forward simulation code from earlier
forward_simulate_roll_and_pick <- function() {
  x <- rdunif(1, min=1, max=4) # 1 uniform betwee
  y <- rdunif(1, min=1, max=x)
  c(x, y) # return a vector with these two realiz
}

sum <- 0.0
n_iterations <- 10000
for (iteration in 1:n_iterations) {
  sample <- forward_simulate_roll_and_pick()
  sum <- sum + sample[1]^sample[2]
}
print(sum/n_iterations)
```

等prob. 抽取.

M=10000

$\hat{G}$ 10000.

```
[1] 25.7831
```

Guarantees from the Law of Large Numbers

Question: How can we make $\approx$ more formal in Equation 1?

Proposition (Law of Large Numbers, LLN): if $Z_1, Z_2, \ldots$ are i.i.d. random variables with $\mathbb{E}|Z_i| < \infty$, then[1]

$$\frac{1}{M}\sum_{m=1}^{M} Z_m \to \mathbb{E}[Z_1].$$

Picking $Z_m = G^{(m)}$ we arrive to the following formalization of $\approx$: for any approximation error tolerance, we can find a number of iterations M large enough such that we will be within that error tolerance with high probability after M iterations.

Footnotes

1. Here "$\to$" denotes a suitable notion of convergence of random variables. In STAT 302 you may have seen LLN with "convergence in probability", but this can be strengthen to "convergence almost sure." The difference between these notions of convergence will not matter in this course. ↩